

Educational Software Tools

Discover some of the best educational tools that make the process of teaching and learning easy and interactive. The DVD has some useful software for businesses too



SRINIVAS N.C.

NASA World Wind

World Wind is an Open Source virtual globe developed by NASA and the Open Source community for use on personal computers. It comes in two versions—**.NET** and **Java**.

World Wind **Java** is a software development kit that is aimed at developers and is not a standalone virtual globe app in the style of Google Earth. It includes a suite of basic demo apps. The program overlays NASA and USGS satellite imagery, aerial photography, topographic maps, Keyhole Markup Language (KML) and Collada files.

The software comes with an extensive suite of plugins. Apart from the Earth, there are several worlds in World Wind, including Moon,

Mars, Venus, Jupiter (with the four Galilean moons of Io, Ganymede, Europa and Callisto) and SDSS (imagery of stars and galactics). Users can interact with the selected planet by rotating it, tilting the view, and zooming in and out. Names of five million places, political boundaries, latitude/longitude lines and other data can be displayed.

World Wind.**NET** provides the ability to browse maps and geospatial data on the Internet using OGC's WMS servers (version 1.4 also uses WFS for downloading placenames), and import ESRI shapefiles and kml/kmz files.

Other features of World Wind.**NET** include support for DirectX 3D polygon mesh models and advanced visual effects such as atmospheric scattering or sun shading. World Wind uses digital elevation model

data collected by NASA's Shuttle Radar Topography Mission, National Elevation Dataset, and Advanced Spaceborne Thermal Emission and Reflection Radiometer. This means one can view topographic features such as the Grand Canyon or Mount Everest in three dimensions. In addition, World Wind has bathymetry data, which allows users to see ocean features, such as trenches and ridges, in 3D.

Many people using the applications are adding their own data and making them available through various sources such as World Wind Central. All the images and movies created with World Wind using Blue Marble, Landsat or USGS-public domain data can be freely modified, re-distributed and used on websites, even for commercial purposes.

ATutor

ATutor is an Open Source Web-based learning management system used to develop and deliver online courses. Administrators can install or update ATutor in minutes, develop custom themes to give ATutor a new look, and easily extend its functionality with feature modules. Educators can quickly assemble, package and re-distribute Web-based instructional content, easily import prepackaged content, and conduct their courses online. Students learn in an accessible, adaptive, social learning environment.

ATutor includes a variety of features to ensure that the content is accessible to all the potential users, including those with slow Internet connections and older Web browsers, and people with disabilities using assistive technologies to access the Web.